Splines as KAN activations

Catmull-Rom, B-spline, Chebyshev, the CR–Chebyshev view, and Kochanek-Bartels

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Abstract

In HSiKAN every edge carries a *learnable per-channel univariate function* $\varphi_c : \mathbb{R} \to \mathbb{R}$ — the KAN activation. Which spline realises φ_c sets four things at once: **expressiveness** (can it bend sharply?), **smoothness** (does it regularise?), **parameter efficiency**, and **evaluation cost** (the launch-bound). This note lays out the four bases we use or could use — Catmull-Rom (the current cell), cubic B-spline, Chebyshev, and the new **CR–Chebyshev** hybrid — plus the Kochanek-Bartels generalisation, with the maths, rendered curves, and a comparison table.

1 The KAN-activation setting

A KAN layer replaces fixed scalar activations with learned univariate functions on the edges. Each φ_c is parameterised by a few weights (control points / coefficients) and evaluated on a softly-bounded input $x \in [-1, 1]$. The signed-message-passing of HSiKAN then aggregates φ-transformed features. So the spline is the unit of nonlinearity, and its basis is a genuine design axis.

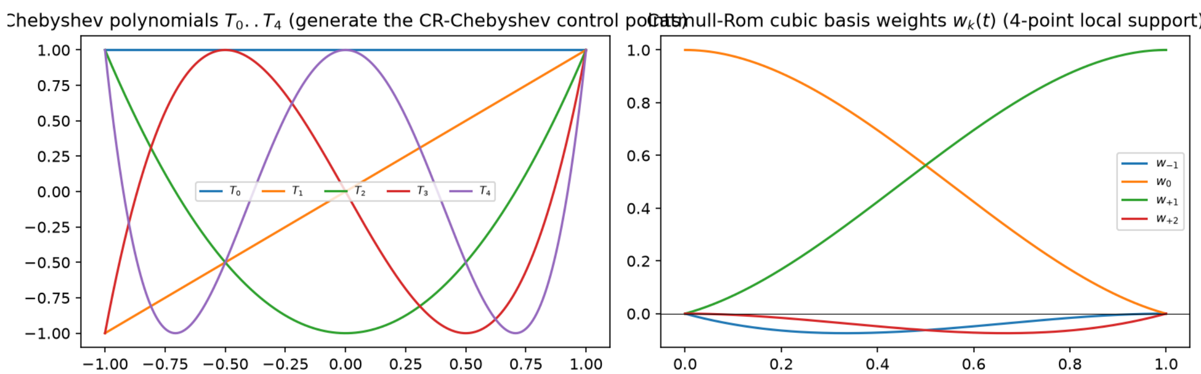


Figure 1: Left: the Chebyshev polynomials $T_0..T_4$ (recurrence $T_{n+1} = 2xT_n - T_{n-1}$) — orthogonal on $[-1, 1]$, global support, the generators of the CR–Chebyshev control points. Right: the Catmull-Rom cubic basis weights $w_k(t)$ — four-point *local* support, summing to one.

2 Catmull-Rom — interpolating, C^1, local (the current cell)

On the segment between control points p_i, p_{i+1} with $t \in [0, 1]$,

$$P(t) = \sum_{k=-1}^{2} w_k(t)\, p_{i+k}, \qquad \begin{aligned} w_{-1} &= \tfrac{1}{2}(-t^3 + 2t^2 - t), & w_0 &= \tfrac{1}{2}(3t^3 - 5t^2 + 2), \\ w_{+1} &= \tfrac{1}{2}(-3t^3 + 4t^2 + t), & w_{+2} &= \tfrac{1}{2}(t^3 - t^2), \end{aligned}$$

equivalently a cubic Hermite with tangents $m_i = \frac{1}{2}(p_{i+1} - p_{i-1})$. It **interpolates** (the curve passes through every p_i), is C^1, has **local** 4-point support, and can overshoot near sharp data. Cost: the four control points are *gathered* per sample — ∼45 small ops, the dispatch-bound core of HSiKAN's forward.

3 Cubic B-spline — approximating, C^2, smoother

$P(t) = \sum_i N_{i,3}(t)\, c_i$ with the Cox–de Boor basis $N_{i,k}$ (recursive in the order k). Unlike CR, the curve does **not** pass through the coefficients c_i — it *approximates*, staying inside their convex hull, and is C^2 (one degree smoother than CR). The coefficients are control *handles*, not curve values.

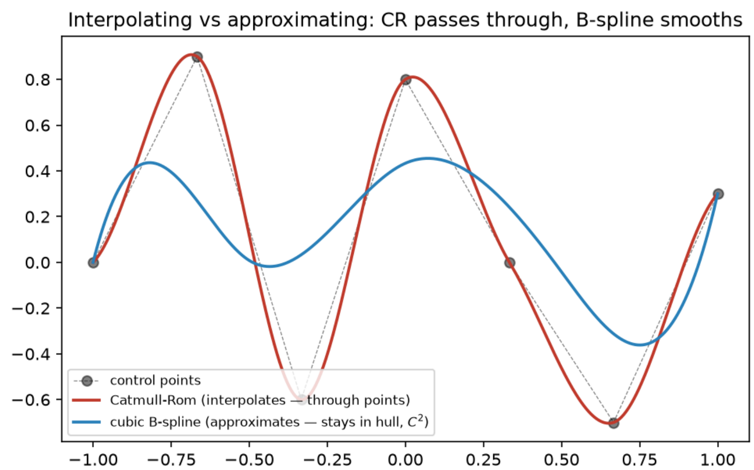


Figure 2: Same control points: Catmull-Rom passes through them (interpolating, can overshoot); the cubic B-spline smooths inside the hull (approximating, C^2).

4 Chebyshev and the CR–Chebyshev view

Chebyshev: $T_0{=}1$, $T_1{=}x$, $T_{n+1}{=}2xT_n - T_{n-1}$ — orthogonal polynomials with the *minimax* (near-optimal uniform) approximation property, **global** support, evaluated as a single Vandermonde *matmul* (no gather, no branching) so they are the cheapest basis at $B{=}1$. A function is written $f(x) \approx \sum_k c_k T_k(x)$.

CR–Chebyshev (the new view): let the CR control points be *generated* by a degree-$(k{-}1)$ Chebyshev expansion sampled at the fixed knots,

$$P \;=\; C\, T_{\text{knots}}^{\top}, \qquad T_{\text{knots}} \in \mathbb{R}^{G\times k} \text{ (cached constant)},$$

with $C \in \mathbb{R}^{(\text{channels})\times k}$ learned. The control points are now a **band-limited** (smooth) family rather than free knots — this is Catmull-Rom seen *through a Chebyshev lens*. Consequences: k coefficients/channel ($k < G \Rightarrow$ param-efficient), a smoother fit on smooth targets, and — because the function lies on a degree-k Chebyshev — it is Chebyshev-representable, so it can be **trained with the CR gather but deployed with the cheap Chebyshev matmul** (∼3× at $B{=}1$). The honest cost: band-limiting caps sharp/narrow features; k is the knob.

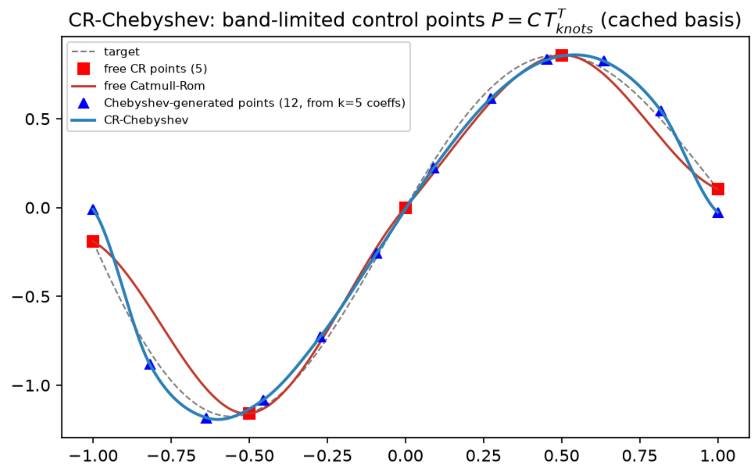


Figure 3: CR–Chebyshev: k=5 learned coefficients place G=12 smooth control points (blue), interpolated by CR — closer to the smooth target than 5 free CR points (red), at equal parameter count.

5 Kochanek-Bartels — learnable shape (TCB)

KB is the cubic-Hermite generalisation whose tangents carry **Tension, Continuity, Bias**:

$$m_i^{\text{out}} = \tfrac{(1-T)(1+B)(1+C)}{2}\Delta_i^- + \tfrac{(1-T)(1-B)(1-C)}{2}\Delta_i^+, \quad m_i^{\text{in}} = \tfrac{(1-T)(1+B)(1-C)}{2}\Delta_i^- + \tfrac{(1-T)(1-B)(1+C)}{2}\Delta_i^+,$$

with $\Delta_i^- = p_i - p_{i-1}$, $\Delta_i^+ = p_{i+1} - p_i$. At T=C=B=0 this *is* Catmull-Rom. Made **learnable** (per channel, or three global scalars), TCB adds shape degrees of freedom CR cannot express: tension trades overshoot↔taut, continuity sets corner sharpness, bias sets asymmetry. It is **orthogonal** to CR–Chebyshev (TCB shapes the *tangents*; Chebyshev shapes the *points*), so the richest cell is "Chebyshev points + learnable TCB tangents."

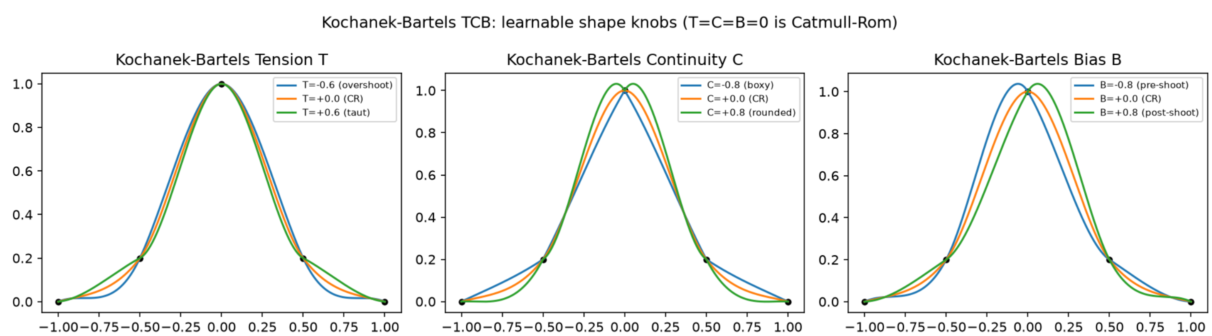
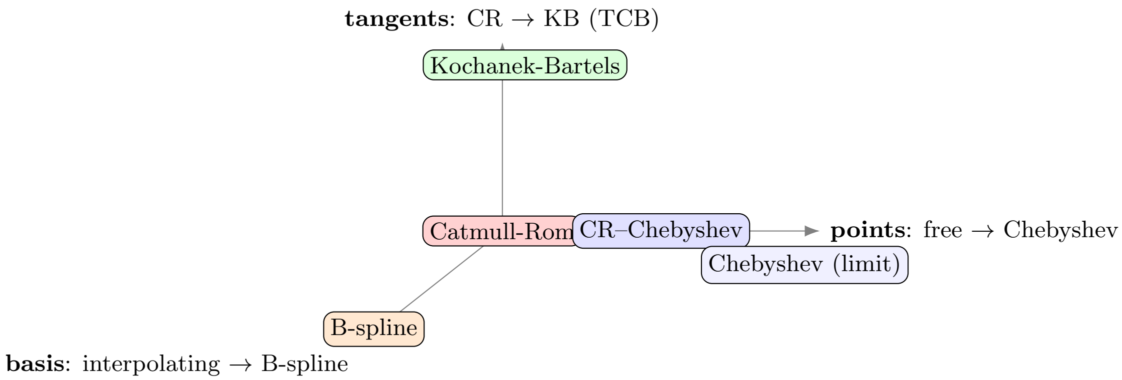


Figure 4: The three Kochanek-Bartels knobs on fixed control points. Tension tightens/overshoots; continuity rounds or sharpens corners; bias shifts the curve before/after each point. T=C=B=0 recovers Catmull-Rom.

6 The 3-axis design space — Catmull-Rom at the origin

The four constructions are not a list but a **basis for a design space**: Catmull-Rom sits at the origin and each generalisation moves along *one* axis.



Axis 1 — control points (a smoothness prior on the knots): free points → band-limited Chebyshev expansion. More Chebyshev = fewer parameters, smoother, cheaper, but a lower fit ceiling; the *pure* Chebyshev (no gather) is the global limit. **Axis 2 — tangents:** fixed Catmull-Rom → learnable Kochanek-Bartels TCB — the same fit on these targets but extra shape degrees of freedom (overshoot, corners, bias). **Axis 3 — basis mode:** interpolating Hermite (CR/KB pass through the points) → approximating B-spline (C^2, stays in the hull). They **compose** (e.g. Chebyshev points + TCB tangents); axes 1–2 are orthogonal *within* the interpolating family, while axis 3 switches the basis itself (a B-spline has no explicit tangents, so it does not carry axis 2). **So B-splines are not omitted — they are the third axis.**

basis	smooth	bump	sharp	B=1 μs	B=256 μs	params
Catmull-Rom (origin)	0.0001	0.0004	0.0000	656	802	8
B-spline (axis 3)	0.0000	0.0001	0.0001	757	1236	10
Chebyshev (axis 1 limit)	0.0004	0.0004	0.0022	**319**	**389**	8
CR–Chebyshev (axis 1)	0.0039	0.0104	0.0350	442	476	**5**
Kochanek-Bartels (axis 2)	0.0001	0.0003	0.0000	754	741	11

Table 1: Measured: gradient-fit MSE to each target (1 channel, 400 Adam steps) + latency + params. CR / B-spline / KB have the degrees of freedom to fit anything; pure Chebyshev is the fastest but band-limits the *sharp* target; CR–Chebyshev is the param-efficient, smooth, fast *middle* — fewest params, a built-in smoothness *regulariser* — at a deliberately lower fit ceiling; B-spline is C^2-smooth but slowest batched. The cell choice is therefore a point in the cube, picked by what a layer needs: raw local fit (CR), speed (Chebyshev), parameter-efficiency/regularisation (CR–Chebyshev), shape control (KB), or C^2 smoothness (B-spline).

7 Which, for HSiKAN

CR–Chebyshev is the new default cell for SA-HSiKAN (shipped, tested): smoother and param-efficient, with the train-CR / deploy-Chebyshev bridge that helps the launch-bound rollout. **Kochanek-Bartels** is the expressiveness ablation when a channel needs sharp or asymmetric shape. **Chebyshev-**

only is the fast deploy form. Plain **Catmull-Rom** stays the interpolation baseline. All are one `make_activation` call apart (`cr` / `cr_cheby` / `bspline`), so the choice is a config, not a fork.